

Cecilie Glittum

Curriculum Vitae

PERSONAL DETAILS

Born	11 November 1996
Nationality	Norwegian
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POSITIONS

MSCA Postdoctoral Research Fellow (COFUND) 2025-

University of Oslo

Project leader of the DSPINN project, where we investigate the possibilities of stabilising quantum spin liquids from kinetic frustration using DMRG and neural networks.

Postdoctoral Researcher 2024-2025

Helmholtz-Zentrum Berlin für Materialien und Energie

EDUCATION

PhD in Physics 2020-2024

Trinity College, University of Cambridge

Theory of Condensed Matter group at the Cavendish Laboratory.
Thesis title: *Wonderland of frustration: Magnetic moments and itinerant electrons on the pyrochlore lattice.*
Date of defence: 25.10.2024

MSc. in Theoretical Physics 2018-2020

University of Oslo

Thesis title: *Nematic Bond Theory of Frustrated Heisenberg Models on Triangular and Honeycomb Lattices.*
GPA: 5.00 (A)

BSc. in Physics, Astronomy and Meteorology 2015-2018

University of Oslo

Specialisation in physics.
Research project title: *Field Theory of Frustrated Magnets.*
GPA: 4.94 (A)

AWARDS AND FUNDING

- Marie Skłodowska-Curie Actions Postdoctoral Fellowship (COFUND) 2025-2028
- Institute of Physics Sam Edwards PhD Thesis Prize: Theory of Condensed Matter Group 2025, runner-up
- **American Physical Society Outstanding Dissertation in Magnetism Award 2025**
- Martin Landrøs pris for fremragende mastergrad i fysikk 2021 (EN: Martin Landrø's Prize for Outstanding Master Thesis in Physics 2021) awarded by the Norwegian Physical Society
- Aker Scholarship 2020 (4 years of PhD studies in Cambridge fully funded)

SELECTED INVITED TALKS

- **Scheduled: *Kinetic frustration as a new route to quantum matter*, The 14th International Conference on Highly Frustrated Magnetism HFM27, Munich, Germany (July 2027)**
- *RVB physics from kinetic magnetism*, Advanced Working Group on Kinetic Magnetism, Cambridge, UK (2026)
- *Doping-induced quantum spin liquids: Towards robust topological order for quantum technology*, IQT Nordics Junior Researcher Symposium, Oslo (2026)
- *A resonant valence bond spin liquid in the dilute limit of doped frustrated Mott insulators*, Nordic Conference on Recent Trends in Condensed Matter Physics, Niels Bohr Institute, Copenhagen (2025)
- *Doping-induced RVB spin liquid in a frustrated Mott insulator*, University of Oslo (2025)
- *Field-induced Multi-$\vec{Q}$ States in a Pyrochlore Heisenberg Magnet*, TCM seminar, Cavendish Laboratory, University of Cambridge (2025)
- ***Wonderland of frustration: Magnetic moments and itinerant electrons on the pyrochlore lattice*, Award talk at the APS Global Physics Summit (March Meeting), Anaheim (2025)**
- *Doping-induced spin liquid in a pyrochlore magnet*, Smalley-Curl Institute, Rice University (2024)
- *Doping-induced spin liquid in a pyrochlore magnet*, Los Alamos National Laboratory, Los Alamos (2024)
- *Sublattice pairing in pyrochlore Heisenberg antiferromagnets*, KTH Stockholm (2024)

TEACHING EXPERIENCE

Lecturing

University of Oslo

- FYS4480 - *Quantum Mechanics for Many-Particle Systems* (2026): Co-responsible for lecturing, group sessions and organising the course.
- FYS4110 - *Advanced Quantum Mechanics* (2025): Guest lecture on quantum magnetism and magnons.

Supervision

- 3-month summer project for a second-year BSc student at UiO (2026)
  Theme: Numerical study of spin dipole-interactions on the pyrochlore lattice
- Main supervisor of MSc student from École normale supérieure de Lyon (while in Berlin) (2025)
  Title: Study of vortices in XY magnets on the triangular lattice via classical Monte Carlo simulations

Teaching assistant

University of Cambridge

Part II Advanced Quantum Physics (2020), Theoretical Physics I (2020, 2021, 2022, 2023), Part II Computational Physics (2023).

University of Oslo

FYS-MEK 1110 - *Mechanics* (2017, 2019), MAT-INF 1100 - *Modelling and Computations* (2017).

Course material development

University of Cambridge

Constructed the problem set on neural networks for Part II Computational Physics (2023).

Simula Research Laboratory/University of Oslo

Constructed the problem sets used in two courses given at the University of Oslo: IN1900 - *Introduction to Programming with Scientific Applications* and IN1910 - *Programming with Scientific Applications* (2018).

Evaluation of multiple-choice questions in physics examinations

Statistical study of student performance on multiple-choice and free-response questions in three physics examinations at the University of Oslo (2020).

Writing Science instructor

Led sessions on academic writing for first year master's students at the Department of Physics, University of Oslo (2019).

Study programme development

Evaluation of the structure and curriculum of the bachelor's programme in Physics and Astronomy at the University of Oslo (2019).

OTHER ACADEMIC EXPERIENCE

Committees

University of Oslo

- Fixed-term representative on the Faculty Board for The Faculty of Mathematics and Natural Sciences (2026)
- Norwegian Agency for Quality Assurance in Education (NOKUT) evaluation of labour-market relevance in disciplinary programmes (BSc in Physics and Astronomy at the University of Oslo) (2019 - 2020)
- Programme Council of the Bachelor's Programme in Physics and Astronomy (2019 - 2020)
- Programme Council of the Master's Programme in Physics (2018 - 2019)
- The Board of the Department of Physics (deputy) (2017 - 2020)

University of Cambridge

- Secretary and member of the executive committee of Trinity College BA Society (2021-2022)
- Secretary and member of the Liaison Committee, Trinity College (2021-2022)

Referee

APS Physical Review: Letters (1), B (7), Research (4)

Outreach

- Welcome-week seminar for bachelor's students in quantum technology, University of Oslo (2026)
  Title: *Når magneter blir frustrerte: Fra kjøleskapsmagneter til kvantespinnvæsker.*
- Fysikk i Origo (seminar series arranged by the Student Physics Association), University of Oslo (2026)
  Title: *Frustrated magnets and quantum spin liquids.*
- Seminar on academic career for final year master's students in physics, University of Oslo (2026)
  Title: *The Life of a Postdoc.*
- Welcome week seminar for new bachelor's and master's students in physics, University of Oslo (2026)
  Title: *Academic career: How to prepare during your studies.*
- Prepare Project (2016-2019), Simula School of Research and Innovation
  Visiting high schools in the Oslo-region giving presentations about natural sciences and study opportunities, with an aim to motivate and inspire towards studies in STEM.

SKILLS

Languages	Norwegian (Mother tounge), English (Fluent), German (Basic)
Programming	C++, Python, Mathematica
Techniques	Exact diagonalisation, tensor networks/matrix product states, Nematic Bond Theory (field-theoretic framework), classical Monte Carlo